

- 1 The performance of your neural network is disappointing. Which of the following actions is LEAST likely to improve performance?

Select one alternative:

- ☐ All of the options listed here are perfectly reasonable actions to try.
- ☐ Add batch normalization to several of the layers.
- ☐ Increase both the minibatch size and the learning rate.
- ☐ Supplement your input embeddings with attention and positional encoding.
- ☐ Change the activation function from non-linear to linear for all layers. 

Maximum marks: 3

- 2 A standard feed-forward neural network has exactly 5 layers, consisting of 10, 20, 30, 30 and 7 neurons, respectively. Each layer uses a linear activation function. The weight matrix W connects the 20-neuron layer to the 30-neuron layer. A dataset consisting of 100 training cases is run through the network for 50 epochs, using a minibatch size of 25, with each case being fed through the network exactly once per epoch. How many times is weight $W_{7,11}$ modified during this training session?

Select one alternative:

- ☐ 50
- ☐ 120000
- ☐ 100
- ☐ 600
- ☐ 5000
- ☐ 3000
- ☐ 200



Maximum marks: 2

- 3 A standard feed-forward neural network contains many layers, including X and Y, which are linked via weight matrix W . X uses a sigmoid activation function, while Y uses a hyperbolic tangent (tanh).

Which of the following are the first few factors of the gradient that is used to update weight $W_{j,k}$ (the weight on the connection from x_j to y_k)?

Select one alternative:

- ☐ $(1 - x_j)y_k$
- ☐ $(1 - x_j^2)y_k(1 - y_k)$
- ☐ $x_j(1 - x_j)w_{j,k}(1 - y_k^2)$
- ☒ $x_j(1 - y_k^2)$
- ☐ $x_j(1 - x_j)(1 - y_k^2)$
- ☐ $1 - y_k^2$
- ☐ $x_jw_{j,k}(1 - y_k^2)$



Maximum marks: 4

4 A neural network has many layers, including these three hidden layers: X, Y and Z.

X sends its outputs to Y via weights U, and Y sends its outputs to Z via weights V.

You do not know the activation functions used in any of the layers. Which one of the following expressions would you LEAST expect to see as part of a formula for the gradients of this network?

Select one alternative:

- ☐ $v_{k,j}(1 - z_j^2)$
- ☐ $v_{j,k}(1 - z_k^2)$
- ☐ $y_j(1 - v_{j,k})z_k(1 - z_k)$ ✔
- ☐ $u_{j,k}(1 - y_k^2)$
- ☐ $u_{k,j}y_j(1 - y_j)v_{j,k}z_k(1 - z_k)$
- ☐ $u_{j,k}y_k(1 - y_k)$

Maximum marks: 3

- 5 Of the people listed below, who has made the most significant technical contributions to the underlying methods in many of today's most impressive AI systems?

Select one alternative:

- ☐ Allan Newell
- ☐ Aaron Courville
- ☐ Max Tegmark
- ☐ John McCarthy
- ☐ Elon Musk
- ☐ Geoffrey Hinton
- ☐ Fei-Fei Li
- ☐ Bill Gates



Maximum marks: 1

- 6 Several different programmers were given the same text corpus and asked to generate embeddings. You have sampled a few words and their 6-bit embeddings from each person. Which sample is almost certainly from a poorly trained network?

Select one alternative:

- ☐ Beer(000111), Wine(001111), Whiskey(010111)
- ☐ White(001100), Orange(010101), Black(110011)
- ☐ England(000011), Scotland(000001), Argentina(010101)
- ☐ Lion(100101), Cougar(000101), Tiger(100001)
- ☐ Robin(100000), Woodpecker(011101), Rattlesnake(110000) 

Maximum marks: 2

- 7 A neural network has many layers, with the final two being X (with 3 nodes) and Y (with one node). Three weights ($w_1 = 0.2$, $w_2 = 0.7$ and $w_3 = -0.8$) connect x_1, x_2, x_3 to y_1 .

On the most recent forward pass through the network, the outputted values of X are:

$$x_1 = 0.5, x_2 = 1.0, x_3 = -0.25$$

The activation function for node y_1 is $y_1 = 2s^3$, where s is the sum of weighted inputs into y_1 .

The error function is $(T - y_1)^2$, where T is the target value. In this case, T = 1.

The learning rate is 0.1, and the minibatch size is 1. Find the new values for the three weights after the backward pass of backpropagation. (All answers must include at least one digit before the decimal point, even if it is a zero, e.g. 0.3 or -0.9)

$$w_1 = \boxed{} (-0.4) \quad w_2 = \boxed{} (-0.5) \quad w_3 = \boxed{} (-0.5)$$

Maximum marks: 4

8 Assume that the loss function for a neural network is:

$$L = E + \alpha \Omega(\theta)$$

where E is the cross-entropy error and $\Omega(\theta)$ is a parameter regularizer, with θ representing the collection of all weights and biases; α is a constant.

It is unknown whether Ω is an L1 or an L2 regularizer.

Select all of the alternatives below that are true:

- ☐ If Ω is L1, then $\frac{\partial \Omega(\theta)}{\partial w_i}$ can have only 3 possible values, regardless of the value of w_i ✓
- ☐ If Ω is L2, then $\alpha \Omega(\theta)$ can never exceed π^3
- ☐ If Ω is L2, then learning tends to be less stable than if it were L1.
- ☐ If the absolute values of weights are small, then Ω will have a greater effect upon the loss if it is L1 than if it is L2. ✓

Maximum marks: 3

9 If a neural network outputs these 8 probabilities:

$1/32, 1/8, 1/32, 1/8, 3/32, 1/16, 1/2, 1/32$

and has these corresponding targets:

$1/8, 0, 1/8, 0, 0, 1/2, 1/4, 0$

Compute the cross-entropy error. Calculate all logarithms base 2, (e.g. $\log_2(8) = 3$), and express your answer as a decimal number with one digit after the separator (e.g. 1.7) .

Your answer: (3.5) .

Maximum marks: 3

10 Which of the following are true statements about the sigmoid activation function:

Select one or more alternatives:

- ☐ It only outputs negative values when given very large negative inputs.
- ☐ If Q is the sum of weighted inputs to a sigmoid, then the derivative of that sigmoid is $Q(1-Q)$.
- ☐ The RELU activation function has become much more popular than the sigmoid. ✓
- ☐ The relatively flat behavior on both ends of the sigmoid curve is a major contributor 'vanishing gradients'. ✓

Maximum marks: 3

- 11 A basic feedforward neural net has 3 layers: X (input), Y (hidden) and Z (output), with weight matrix U connecting X to Y, and weight matrix V connecting Y to Z.

Each column of a weight matrix houses the weights on all of the incoming connections to the receiving layer, and each row houses the weights of all outgoing connections from the sending layer.

The values of the input nodes are represented as a column vector.

The activation functions for layers Y and Z are F_Y and F_Z , respectively.

Layers Y and Z both use biases stored in column vectors B_Y and B_Z , respectively.

Which of the following expressions best represents the matrix operations needed to perform a full forward pass, from layer X to layer Z (where " $\cdot$ " denotes vector-matrix multiplication, and "T" denotes the matrix or vector transpose operation)?

Select one alternative:

- ☐ $F_Z(B_Z^T + (F_Y(X \cdot U + B_Y^T)) \cdot V)$
- ☐ $F_Z(B_Z^T + (F_Y(X^T \cdot U^T + B_Y^T))^T \cdot V^T)$
- ☐ $F_Z((F_Y((X + B_Y^T) \cdot U^T) + B_Z^T) \cdot V^T)$
- ☒ $F_Z(B_Z^T + F_Y(X^T \cdot U + B_Y^T) \cdot V)$
- ☐ $F_Z(B_Z^T + (F_Y(X \cdot U^T + B_Y^T)) \cdot V^T)$
- ☐ $F_Z(V \cdot (F_Y(U \cdot (X + B_Y^T) + B_Z^T))$



Maximum marks: 2

- 12** A neural network has input layer X, which consists of 5 feature maps (a.k.a. channels), each of size 18×18 . Layer X feeds directly into layer Y, which is a convolution layer with 6 feature maps, each of which uses a stack (of depth 5) of 4×4 kernels. Y uses a vertical and horizontal stride length of 3 and does not use zero filling.

How many total neurons are in layer Y ?

Your answer: (150)

Maximum marks: 2

Note: It is easy to count wrong here. If the feature maps were 19×19 , the total number of neurons would be 216 instead of 150.

Both answers were accepted as correct on the exam.

- 13** Recurrent Networks vary in their connection topologies and the manner in which layers update state information. Below are 5 different update formulas for the hidden layer (H) in a network with input layer X, output layer Y and the single hidden layer H. In addition, T represents a target output vector. F_h is the activation function for layer H, and all other subscripts denote timestamps. U is the weight matrix connecting X to H, and V is the weight matrix for the recurrent connections. The input layer of V depends upon the network type.

A: $H_{t+1} = F_h(X_{t+1} \cdot U + T_t \cdot V)$

B: $H_{t+1} = F_h(X_{t+1} \cdot U + H_t \cdot V)$

C: $H_{t+1} = F_h(X_t \cdot V + X_{t+1} \cdot U)$

D: $H_{t+1} = F_h(X_{t+1} \cdot U + Y_t \cdot V)$

E: $H_{t+1} = T_t \cdot V + F_h(X_{t+1} \cdot U)$

For each of the networks below, choose one of A-E to indicate the hidden-layer update that is most appropriate:

1. An Elman Net during training: (B)

2. A Teacher-forced Jordan Net during training: (A)

3. A Teacher-forced Jordan Net during testing: (D)

Maximum marks: 5

- 14 Given the following query, keys and values, compute the embedding vector V^* of length 2 using the basic algorithm for scaled dot-product attention.

Query = [0.4, 0.2, 0.4]

Keys = [0.1, 0.3, 0.6]
[0.6, 0.3, 0.1]

Values = [0.1, 0.9]
[0.7, 0.3]

$V^*[1] =$ (0.4)

$V^*[2] =$ (0.6)

Maximum marks: 4

Note: Depending on whether you interpret the values as being paired by row or column, you will end up with either [0.4, 0.6] or [0.5, 0.5].

Both answers were accepted as correct on the exam.

15

The motivation for adding positional encodings into Transformers is the simple fact that longitude and latitude enrich the context of all spatiotemporal information.

Select an alternative

☐ True

☐ False



Multi-head attention often appears in both the Encoder and Decoder portion of a Transformer.

Select an alternative

☐ False

☐ True



ChatGPT wrote this entire exam

Select one alternative:

☐ True

☐ False



Masked self-attention is much more likely to occur in the Decoder than the Encoder of a Transformer.

Select an alternative

☐ True

☐ False



Any given transformer can have regular attention or self-attention, but not both.

Select an alternative

☐ True

☐ False



Maximum marks: 5

16 Which of the following statements are true about this equation:

$$Z(i, j, k) = (X * w)(i, j, k) = \sum_{\alpha=-a}^a \sum_{\beta=-b}^b \sum_{\gamma=-c}^c X(i + \alpha, j + \beta, k + \gamma) * w(\alpha, \beta, \gamma)$$

Select one or more alternatives:

- ☐ From this equation, one can determine whether zero-padding is necessary.
- ☐ To accomodate all dimensions of the kernel during backpropagation, the gradients will include the matrix product of a Jacobian and a Hessian.
- ☐ The equation gives no indication of the sizes of the strides. ✓
- ☐ For w to work properly, the operations of this equation must be followed by a max-pooling layer.
- ☐ This is the essence of 3-d convolution. ✓
- ☐ The number of weights in the kernel is $(2a + 1) * (2b + 1) * (2c + 1)$ ✓

Maximum marks: 3

17 E1: $R_{t+1} = \rho R_t + (1 - \rho) G_{t+1} \odot G_{t+1}$

E2: $\lambda_{t+1} = \lambda_t \oslash (\delta + R_{t+1})^{1/2}$

E3: $\Delta W_{t+1} = -\lambda_{t+1} \odot G_{t+1}$

where:

- $\odot$ denotes element-wise multiplication,
- $\oslash$ denotes element-wise division,
- ρ is a constant fraction
- δ is a small constant
- all subscripts denote steps in the process
- G is an array of all gradients in the neural network, as calculated by backpropagation.
- R is the same size as G, and R_0 is all zeros
- W and λ are the same size as G.
- λ_0 is an array of equal, non-zero values.

Equations E1-E3 are the basic framework for RMSProp.

Select all of the following that are true statements:

- ☐ The concept of momentum is implicitly handled by R. ✓
- ☐ When any single gradient has had a large absolute value over several steps, E2 will tend to decrease its corresponding learning rate. ✓
- ☐ To base weight change more strongly on more recent gradients, adjust ρ ✓
- ☐ RMSProp is a popular regularizer.
- ☐ This procedure works well for classification but poorly for regression tasks.
- ☐ RMSProp is a special instance of Batch Normalization

Maximum marks: 5

- 18 A simple neural network has the following topology: $\mathbf{X} \rightarrow \mathbf{Y} \rightarrow \mathbf{Z}$. The weight matrices are V (for $\mathbf{X} \rightarrow \mathbf{Y}$) and W (for $\mathbf{Y} \rightarrow \mathbf{Z}$). Layers X and Y contain hundreds of neurons each, and each of those neurons uses a sigmoid activation function, while layer Z consists of a single neuron (z_0) that uses this linear activation function:

$$F_z(s) = 2s$$

where s is the sum of weighted inputs into z_0 .

L is the loss function, and $\delta_z = \frac{\partial L}{\partial z_0}$

Consider the following expression:

$$x_4 y_9 (1 - y_9) \sum_k 2w_{k,0} \delta_z$$

This is the entry in a Jacobian Matrix. Which of the options below best represents that matrix.

Select one alternative:

☐ $J_Y^W J_W^Z J_Z^L$

☐ $J_X^Y J_Y^W J_W^Z J_Z^L$

☐ $J_X^Y J_Y^Z$

☐ $J_V^Y J_Y^Z J_Z^L$



☐ $J_X^Y J_Y^Z J_Z^L$

☐ $J_X^Y J_Y^Z J_Z^L$

Maximum marks: 3

19 Which of the following is a true statement?

Select one or more alternatives:

- ☐ If a neural network for written character recognition is trained on cases of only right-handed students but is then tested on data that includes some left-handed students, the use of Batch Normalization in several layers is a good choice. ✓
- ☐ The use of LSTMs can help reduce the problem of vanishing gradients. ✓
- ☐ When the neural network outputs a probability distribution, one of the best loss functions to use is Mean Squared Error (MSE).
- ☐ Transformers rely heavily on backpropagation through time (BPTT)
- ☐ Neural network research was stalled for many years in the 1970's and 1980's due to very simple problems with primitive implementations of matrix multiplications.

Maximum marks: 3

- 20** You built a simple neural network to handle the 4 exclusive-or (XOR) cases: the true cases (0,1) and (1,0), and the false cases (0,0), (1,1). The net has an input layer with two nodes and it connects directly to an output layer with one node, which uses a step activation function. Since XOR is not linearly separable, this network cannot learn to classify these four cases with perfect training accuracy, even when using backpropagation. Which of the enhancements below would enable backpropagation to achieve perfect training accuracy?

Select one or more alternatives:

- ☐ Add a second weight vector connecting the input to the output layer.
- ☐ Add a recurrent weight vector that connects the input layer to itself.
- ☐ Change the activation function of the output layer to a hyperbolic tangent.
- ☐ Add a hidden layer with two neurons, each of which uses a sigmoid activation func: ✓ .
- ☐ Add a hidden layer with two neurons, each of which uses a linear activation function.

Maximum marks: 3

- 21 Variational auto-encoders (VAEs) extend ordinary auto-encoders (AEs) by assuming that the "code" (activation out of the bottleneck-layer) follows a standard Gaussian distribution. Which of the following is the **least** correct (or indeed the incorrect) claim regarding the consequences of this model extension?

Select one alternative:

- ☒ The VAE can be used for **classification** and produce good results. The results from an AE are typically *much* poorer.
- ☐ The VAE can be used for **anomaly detection** and produce good results. The results from an AE are typically *much* poorer.
- ☐ The VAE can be used as a **generative model** and produce good results. The results from an AE are typically *much* poorer.

Maximum marks: 3

- 22 What are the consequences of using the KL from q to p , i.e., $D_{KL}(q||p)$ as our objective to optimize q ("information projection") instead of the KL divergence from p to q ("moment projection") when doing variational inference?

For simplicity we assume that p is a Mixture of univariate Gaussian distributions, while the approximation q is assumed to be a single univariate Gaussian with unknown parameters μ and σ^2 when declaring the alternatives below.

Select one or more alternatives:

- ☐ None of the other alternatives are guaranteed to be correct, but will depend on the true distribution p that we try to approximate.
- ☐ The variance in the solution (distribution q) will over-estimate the variance of the target-distribution p , meaning that the estimated σ^2 is larger than or equal to the true variance of p .
- ☐ The variance in the solution (distribution q) will under-estimate the variance of the target-distribution p , meaning that the estimated σ^2 is smaller than or equal to the true variance of p . ✓
- ☐ This KL divergence $D_{KL}(q||p)$ is a lower-bound for the log likelihood of the data, while the alternative $D_{KL}(p||q)$ is not.
- ☐ The estimated mean will underestimate the mean of p , in fact it holds that $\mathbb{E}_p[X] \geq \mu + \sigma^2$, where μ and σ^2 are the estimated mean and variance, respectively.

Maximum marks: 3

23 Which of the statements below is likely to raise the most important challenge if you want to use out-of-the-box pre-trained CLIP-models as a means to implement classifier guidance for text-prompting of diffusion models?

Select one alternative:

- ☐ CLIP defines a shared semantic space for text and images. This is not meaningful for text-*conditioning*, and hence CLIP is not really suitable for guidance.
- ☐ The cosine distance measure typically used to compare text and image embeddings take values in the range $[-1, +1]$. This is not the range used by the classifier in the original development of the algorithm, so the mathematics must be adapted.
- ☐ The original text embedding model is not trained with short descriptive texts, and must be adapted.
- ☐ The hyper-parameter that decides to what extent the prompt $\mathbf{y}$ has an effect on the reverse distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$ can be hard to optimize.
- ☐ The original image embedding model is not trained using blurred images, and must be adapted. ✓

Maximum marks: 3

- 24** Diffusion models can be considered as "stacked" variational auto-encoders (VAEs). Please consider the statements below and select those that are true (if any).

Select one or more alternatives:

- ☐ In the VAE one typically learns both the forward and the reverse. In diffusion models one typically only learn the reverse. ✓
- ☐ The diffusion models have no hyper-parameters, while this is not the case for the VAEs.
- ☐ All latent variables in a diffusion model follow the Gaussian distribution. This differs from the VAEs, where the distribution over the latent variables (the "code") typically is chosen more broadly.
- ☐ Each latent "layer" in the diffusion model has the same size as the input-space. This differs from VAEs, where one typically uses a smaller latent space. ✓

Maximum marks: 3

25 What is the consequence of the so-called "mean field" assumption often used in variational inference?

Select one alternative:

- ☐ The *mean* of each variable, $\mathbb{E}_q(\mathbf{Z}_i)$ is assumed to follow a Gaussian *field*.
- ☒ Each variable z_i is assumed to be *independent* in the approximation $q(\mathbf{z})$. 
- ☐ The *mean* of the parameters of the distribution $q(\mathbf{z})$ is set to zero.

Maximum marks: 3

26 The main computational benefit from maximizing the ELBO, defined as

$\mathcal{L}(q) = \mathbb{E}_{\mathbf{z} \sim q} \left[\log \frac{p(\mathbf{z}, \mathbf{x})}{q(\mathbf{z})} \right]$, instead of minimizing the KL-divergence $D_{KL}(q(\mathbf{z}) || p(\mathbf{z}|\mathbf{x}))$, when doing variational inference is

Select one alternative:

- ☒ The ELBO uses the joint distribution $p(\mathbf{z}, \mathbf{x})$, which is more easily available than the conditional $p(\mathbf{z}|\mathbf{x})$ that is used in the KL calculation. ✓
- ☐ The log-term in the ELBO makes the calculations numerically stable. This is particularly important when $(\mathbf{z}, \mathbf{x})$ is high-dimensional.
- ☐ The expectation in the ELBO can easily be approximated using sampling (so-called Monte Carlo estimation).
- ☐ The calculation of the KL-divergence is computationally unstable. If we want to approximate it by sampling we have to use very many samples to produce reliable results.

Maximum marks: 3

27 What are the main motivations for using XAI models?

Select one or more alternatives:

- ☐ It may become legally required by forthcoming rules and regulations ✓
- ☐ XAI-models outperform traditional methods on skewed datasets
- ☐ XAI-models are faster than deep neural networks, and therefore better for real-time applications
- ☐ Explanations may improve end-user acceptance of model output ✓

Maximum marks: 3

28 There currently is no universal solution to XAI. Different user types require different types of explanations.

Select one alternative:

☐ True



☐ False

Maximum marks: 1

29 Which of the following are popular approaches to explaining machine learning models?

Select one or more alternatives:

- ☐ Shapley trees
- ☐ Adversarial testing (testing model output using GANs)
- ☐ Black box enlightenment
- ☐ Surrogate model explanations ✓
- ☐ Extrinsic explanations ✓
- ☐ Intrinsic explanations ✓

Maximum marks: 3

30 Categorize the different XAI methods according to category**Please match the values:**

	Adversarial testing with GANs	Not an XAI method	Extrinsic explanations	Surrogate model explanations	Black box enlightenment	Intrinsic explanations
LIME	☐	☐	☐	☐ ✓	☐	☐
Model trees	☐	☐	☐	☐ ✓	☐	☐
Saliency maps	☐	☐	☐	☐	☐	☐
SHAP	☐	☐	☐ ✓	☐	☐	☐
Concept detection	☐	☐	☐	☐	☐	☐

Maximum marks: 5

31

The upcoming European AI Act is the world's first AI-specific regulation

Select one alternative:

☐ False

☐ True



The AI Act will feature a risk-based approach

Select an alternative

☐ True



☐ False

Certain AI systems will be banned in the AI Act due to their high risk

Select an alternative

☐ True



☐ False

Certain AI systems will be unregulated in the AI Act due to their low risk

Select an alternative

☐ True



☐ False

Maximum marks: 4

32 The following is the expression for the Shapley value of player i :

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n-|S|-1)!}{n!} (v(S \cup \{i\}) - v(S))$$

Given the following values of the characteristic function for the three players $N = \{1, 2, 3\}$,

$$v(S) = \begin{cases} 1 & \text{if } S \in \{\{1, 3\}, \{2, 3\}, \{1, 2, 3\}\} \\ 0 & \text{otherwise} \end{cases}$$

calculate the Shapley values ϕ_1, ϕ_2, ϕ_3 (expressed as fractions, for example 1/3). Make the fractions as simple as possible, so use, e.g., 1/2 instead of 2/4 or 3/6.

$$\phi_1 = \boxed{} (1) / \boxed{} (6)$$

$$\phi_2 = \boxed{} (1) / \boxed{} (6)$$

$$\phi_3 = \boxed{} (2) / \boxed{} (3)$$

Maximum marks: 3